

G. Narayanamma Institute Technology and Science (For Women)
DSUR Lab Record
IV-I-CSE-A (2025-2026)

Aim: Correlation and Covariance:

a) Develop a program to find the correlation matrix on iris data.

Code:

```
# Load the iris dataset
data(iris)

# Select only the numeric columns for correlation analysis
iris_numeric <- iris[, 1:4]

cor_matrix <- cor(iris_numeric)
print(cor_matrix)
```

Output:

```
data(iris)
> # Select only the numeric columns for correlation analysis
> iris_numeric <- iris[, 1:4]
> # Compute the correlation matrix
> cor_matrix <- cor(iris_numeric)
> # Print the correlation matrix
> print(cor_matrix)
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width
Sepal.Length	1.0000000	-0.1175698	0.8717538	0.8179411
Sepal.Width	-0.1175698	1.0000000	-0.4284401	-0.3661259
Petal.Length	0.8717538	-0.4284401	1.0000000	0.9628654
Petal.Width	0.8179411	-0.3661259	0.9628654	1.0000000

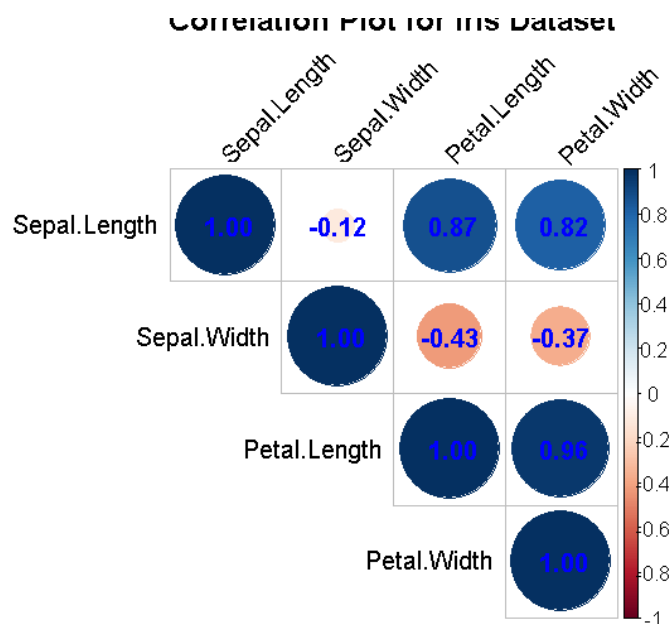
G. Narayanamma Institute Technology and Science (For Women)
DSUR Lab Record
IV-I-CSE-A (2025-2026)

Aim: b) Plot the correlation plot on dataset and visualize giving an overview of relationships among data on iris data

Code:

```
install.packages("corrplot")  
library(corrplot)  
cor_matrix <- cor(iris[, 1:4])  
corrplot(cor_matrix, method="circle", type="upper", tl.col="black", tl.srt=45,  
         addCoef.col="blue", title="Correlation Plot for Iris Dataset")  
  
title = "Correlation Plot for Iris Dataset"  
)
```

Output:



Aim: c) Analysis of covariance: variance (ANOVA), if data have categorical variables on iris data.

Code:

```
# Load the iris dataset
data(iris)

# Perform ANOVA to test if Sepal.Length differs by Species
anova_result <- aov(Sepal.Length ~ Species, data=iris)

# Display the summary of the ANOVA
summary(anova_result)
```

Output:

```
data(iris)
> # Perform ANOVA to test if Sepal.Length differs by Species
> anova_result <- aov(Sepal.Length ~ Species, data=iris)
> # Display the summary of the ANOVA
> summary(anova_result)
```

```
      Df Sum Sq Mean Sq F value Pr(>F)
Species    2  63.21  31.606   119.3 <2e-16 ***
Residuals 147   38.96   0.265
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Aim: a) Import a data from web storage. Name the dataset and now do Logistic Regression to find out relation between variables that are affecting the admission of a student in an institute based on his or her GRE score, GPA obtained and rank of the student. Also check if the model is fit or not. Require (foreign), require (MASS).

Code:

```
require(foreign)
require(MASS)

url <- "https://stats.idre.ucla.edu/stat/data/binary.csv"
admission_data <- read.csv(url)

admission_data$rank <- as.factor(admission_data$rank)
head(admission_data)

#Split into training and testing sets (70/30)
set.seed(123)
sample_index <- sample(1:nrow(admission_data), 0.7 * nrow(admission_data))
train <- admission_data[sample_index, ]
test <- admission_data[-sample_index, ]

# Step 5: Build logistic regression model
log_model <- glm(admit ~ gre + gpa + rank, data = train, family = binomial)

summary(log_model)

#Check model fit using deviance test
p_value <- with(log_model, pchisq(null.deviance - deviance,
                                df.null - df.residual,
                                lower.tail = FALSE))
cat("\nModel Fit (Deviance Test) p-value:", p_value, "\n")

#Predict on test data
predicted_prob <- predict(log_model, newdata = test, type = "response")
predicted_class <- ifelse(predicted_prob > 0.5, 1, 0)

cat("\nConfusion Matrix:\n")
print(table(Predicted = predicted_class, Actual = test$admit))

accuracy <- mean(predicted_class == test$admit)
cat("\nModel Accuracy:", round(accuracy * 100, 2), "%\n")
```

G. Narayanamma Institute Technology and Science (For Women)
DSUR Lab Record
IV-I-CSE-A (2025-2026)

Output:

Loading required package: foreign

Loading required package: MASS

```
A data.frame: 6 × 4
  admit gre gpa rank
  <int> <int> <dbl> <fct>
1     0  380  3.61     3
2     1  660  3.67     3
3     1  800  4.00     1
4     1  640  3.19     4
5     0  520  2.93     4
6     1  760  3.00     2
```

Call:

```
glm(formula = admit ~ gre + gpa + rank, family = binomial, data = train)
```

Coefficients:

```
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -2.821183   1.360473  -2.074  0.03811 *
gre           0.001965   0.001330   1.478  0.13950
gpa           0.650013   0.388017   1.675  0.09389 .
rank2        -1.094622   0.394695  -2.773  0.00555 **
rank3        -1.703651   0.424508  -4.013 5.99e-05 ***
rank4        -1.964562   0.503683  -3.900 9.60e-05 ***
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

```
Null deviance: 360.03  on 279  degrees of freedom
Residual deviance: 327.42  on 274  degrees of freedom
AIC: 339.42
```

Number of Fisher Scoring iterations: 4

Model Fit (Deviance Test) p-value: 4.498297e-06

Confusion Matrix:

```
      Actual
Predicted 0  1
         0 76 22
         1 13  9
```

Model Accuracy: 70.83 %

G. Narayanamma Institute Technology and Science (For Women)
DSUR Lab Record
IV-I-CSE-A (2025-2026)

Aim: b) Apply regression Model to predict the data on any dataset.

Code:

```
require(MASS)
data(mtcars)
head(mtcars)

set.seed(123)
sample_index <- sample(1:nrow(mtcars), 0.7 * nrow(mtcars))
train <- mtcars[sample_index, ]
test <- mtcars[-sample_index, ]

#Robust Regression Model
robust_model <- rlm(mpg ~ wt + hp + cyl, data = train)

summary(robust_model)

predictions <- predict(robust_model, newdata = test)

#Compare actual vs predicted
results <- data.frame(Actual = test$mpg, Predicted = predictions)
print(results)

mse <- mean((results$Actual - results$Predicted)^2)
cat("\nMean Squared Error (MSE):", round(mse, 3), "\n")
```

Output:

G. Narayanamma Institute Technology and Science (For Women)
DSUR Lab Record
IV-I-CSE-A (2025-2026)

A data.frame: 6 × 11

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1
Hornet Sportabout	18.7	8	360	175	3.15	3.440	17.02	0	0	3	2
Valiant	18.1	6	225	105	2.76	3.460	20.22	1	0	3	1

Call: rlm(formula = mpg ~ wt + hp + cyl, data = train)

Residuals:

Min	1Q	Median	3Q	Max
-3.4116	-0.9555	-0.4447	1.2038	5.7158

Coefficients:

	Value	Std. Error	t value
(Intercept)	40.1661	1.7878	22.4661
wt	-3.3902	0.7576	-4.4750
hp	-0.0117	0.0119	-0.9819
cyl	-1.2496	0.5704	-2.1909

Residual standard error: 1.7 on 18 degrees of freedom

	Actual	Predicted
Mazda RX4	21.0	22.495928
Mazda RX4 Wag	21.0	21.631420
Hornet 4 Drive	21.4	20.478744
Valiant	18.1	19.706777
Merc 450SE	16.4	14.259919
Merc 450SL	17.3	15.412596
Lincoln Continental	10.4	9.259084
Toyota Corona	21.5	25.673115
Camaro Z28	13.3	14.277371
Pontiac Firebird	19.2	15.081359

Mean Squared Error (MSE): 5.084

G. Narayanamma Institute Technology and Science (For Women)
DSUR Lab Record
IV-I-CSE-A (2025-2026)

Aim: a) Develop Multiple Regression model for any dataset

Code:

```
data(mtcars) # Load built-in dataset

# Build a multiple regression model
multi_model <- lm(mpg ~ wt + hp + disp, data = mtcars)

# Display summary of model
summary(multi_model)
```

Output:

```
Call:
lm(formula = mpg ~ wt + hp + disp, data = mtcars)

Residuals:
    Min       1Q   Median       3Q      Max
-3.891 -1.640 -0.172  1.061  5.861

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  37.105505   2.110815  17.579  < 2e-16 ***
wt           -3.800891   1.066191  -3.565  0.00133 **
hp           -0.031157   0.011436  -2.724  0.01097 *
disp         -0.000937   0.010350  -0.091  0.92851
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.639 on 28 degrees of freedom
Multiple R-squared:  0.8268,    Adjusted R-squared:  0.8083
F-statistic: 44.57 on 3 and 28 DF,  p-value: 8.65e-11
```


G. Narayanamma Institute Technology and Science (For Women)
DSUR Lab Record
IV-I-CSE-A (2025-2026)

Aim: b) Apply Multiple Regression Model to predict the data on any dataset.

Code:

```
new_car <- data.frame(wt = 3.0, hp = 120, disp = 200)
```

```
# Predict using the fitted model
```

```
predicted_mpg <- predict(multi_model, new_car)
```

```
predicted_mpg
```

Output:

```
> # Predict using the fitted model
> predicted_mpg <- predict(multi_model, new_car)
> predicted_mpg
      1
21.77665
> |
```

G. Narayanamma Institute Technology and Science (For Women)
DSUR Lab Record
IV-I-CSE-A (2025-2026)

Aim: a) Install relevant package for classification.

Code:

```
install.packages(c("caret", "e1071", "rpart"))
```

```
library(caret)  
library(e1071)  
library(rpart)
```

Aim: b) Implement and evaluate the performance of various classifiers with different datasets.

Code:

```
data(iris)  
set.seed(123)  
train_index <- createDataPartition(iris$Species, p = 0.7, list = FALSE)  
train <- iris[train_index, ]  
test <- iris[-train_index, ]
```

```
nb_model <- naiveBayes(Species ~ ., data = train)  
nb_pred <- predict(nb_model, test)
```

```
cat("Naive Bayes Results:\n")  
confusionMatrix(nb_pred, test$Species)
```

```
tree_model <- rpart(Species ~ ., data = train)  
tree_pred <- predict(tree_model, test, type = "class")
```

```
cat("\nDecision Tree Results:\n")  
confusionMatrix(tree_pred, test$Species)
```

Output:

Naive Bayes Results:

```
> confusionMatrix(nb_pred, test$Species)
Confusion Matrix and Statistics
```

	Reference		
Prediction	setosa	versicolor	virginica
setosa	15	0	0
versicolor	0	13	2
virginica	0	2	13

Overall Statistics

```
Accuracy : 0.9111
95% CI : (0.7878, 0.9752)
No Information Rate : 0.3333
P-Value [Acc > NIR] : 8.467e-16
```

```
Kappa : 0.8667
```

```
Mcnemar's Test P-Value : NA
```

Statistics by Class:

	Class: setosa	Class: versicolor	Class: virginica
Sensitivity	1.0000	0.8667	0.8667
Specificity	1.0000	0.9333	0.9333
Pos Pred Value	1.0000	0.8667	0.8667
Neg Pred Value	1.0000	0.9333	0.9333
Prevalence	0.3333	0.3333	0.3333
Detection Rate	0.3333	0.2889	0.2889

Decision Tree Results:

```
> confusionMatrix(tree_pred, test$Species)
Confusion Matrix and Statistics
```

	Reference		
Prediction	setosa	versicolor	virginica
setosa	15	0	0
versicolor	0	14	2
virginica	0	1	13

Overall Statistics

```
Accuracy : 0.9333
95% CI : (0.8173, 0.986)
No Information Rate : 0.3333
P-Value [Acc > NIR] : < 2.2e-16
```

```
Kappa : 0.9
```

```
Mcnemar's Test P-Value : NA
```

Statistics by Class:

	Class: setosa	Class: versicolor	Class: virginica
Sensitivity	1.0000	0.9333	0.8667
Specificity	1.0000	0.9333	0.9667
Pos Pred Value	1.0000	0.8750	0.9286
Neg Pred Value	1.0000	0.9655	0.9355

Code:

```
data(iris)
iris_data <- iris[, -5]
set.seed(123)
km_model <- kmeans(iris_data, centers = 3, nstart = 20)
print(km_model)
table(km_model$cluster)
```

Output:

Available components:

```
[1] "cluster"      "Centers"      "totss"        "withinss"     "tot.withinss"
[6] "betweenss"    "size"         "iter"         "ifault"
> table(km_model$cluster)
```

```
1 2 3
50 62 38
> source("~/active-rstudio-document")
K-means clustering with 3 clusters of sizes 50, 62, 38
```

Cluster means:				
	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width
1	5.006000	3.428000	1.462000	0.246000
2	5.901613	2.748387	4.393548	1.433871
3	6.850000	3.073684	5.742105	2.071053

```
Clustering vector:  
[1] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1  
[42] 1 1 1 1 1 1 1 1 1 2 2 3 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2  
[83] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 3 3 3 3 3 3 3 3 3 3 3 3 3 3  
[124] 2 3 3 2 2 3 3 3 3 2 2 3 3 3 2 3 3 3 2 3 3 3 2 3 3 2
```

```
within cluster sum of squares by cluster:
[1] 15.15100 39.82097 23.87947
(between_SS / total_SS = 88.4 %)
```

Available components:

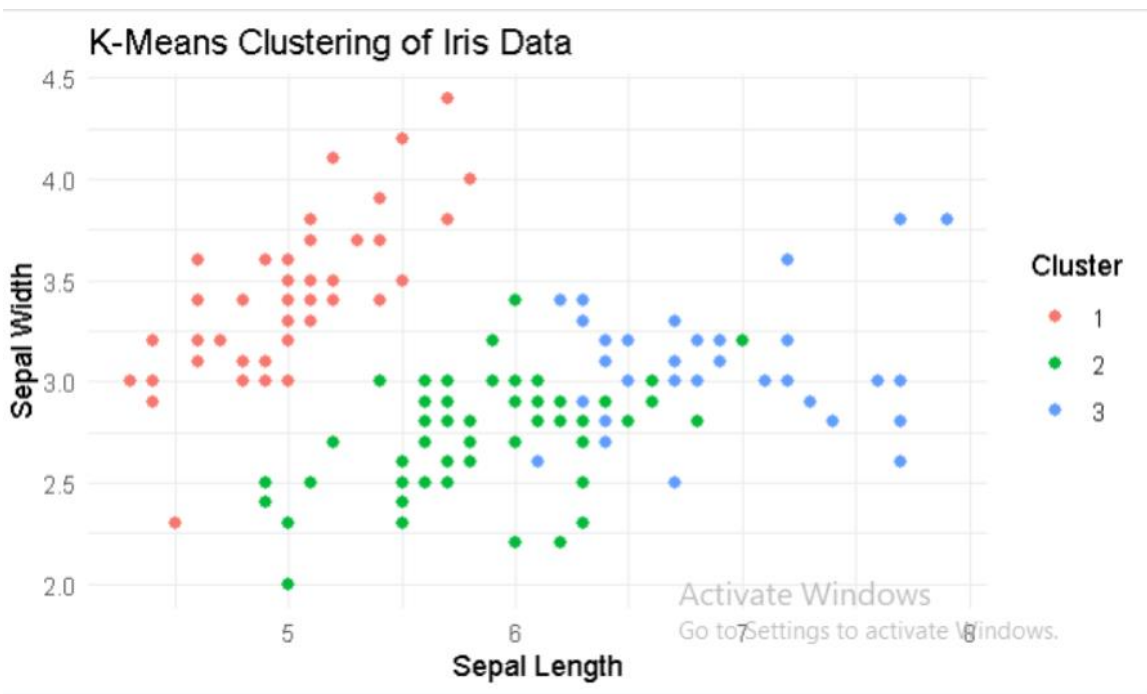
```
[1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
[6] "betweenss"    "size"         "iter"         "ifault"
```

Aim: b) Plot the cluster data using R visualizations.

Code:

```
library(ggplot2)
iris$Cluster <- as.factor(km_model$cluster)
ggplot(iris, aes(x = Sepal.Length, y = Sepal.Width, color = Cluster)) +
  geom_point(size = 2) +
  labs(title = "K-Means Clustering of Iris Data",
       x = "Sepal Length", y = "Sepal Width") +
  theme_minimal()
```

Output:

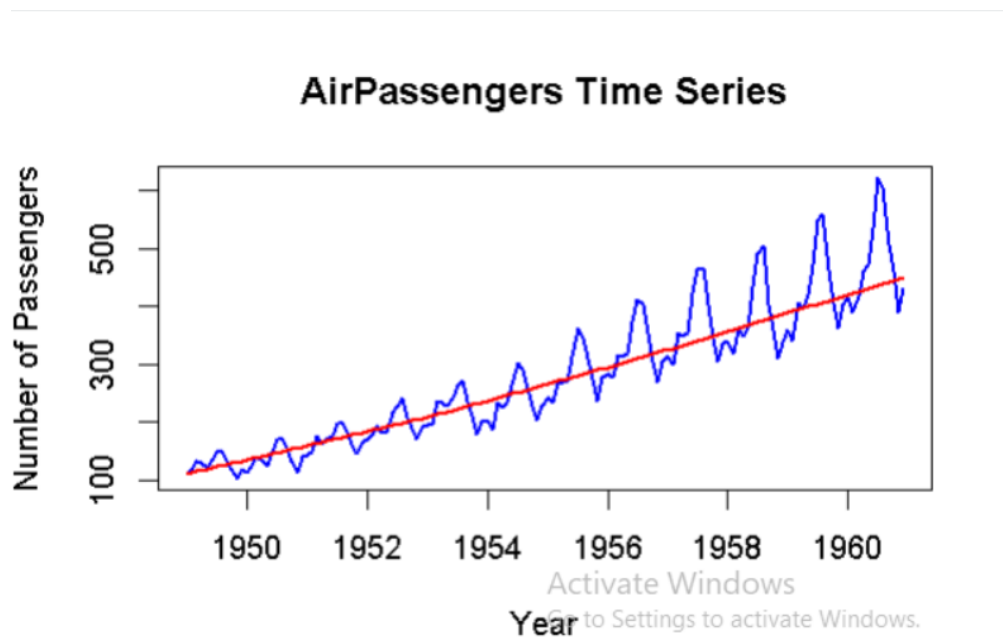


Aim: a) Plot the Time series data using R visualizations.

Code:

```
data(AirPassengers)
ts_data <- AirPassengers
plot(ts_data,
     main = "AirPassengers Time Series",
     ylab = "Number of Passengers",
     xlab = "Year",
     col = "blue",
     lwd = 2)
lines(lowess(time(ts_data), ts_data), col = "red", lwd = 2)
```

Output:



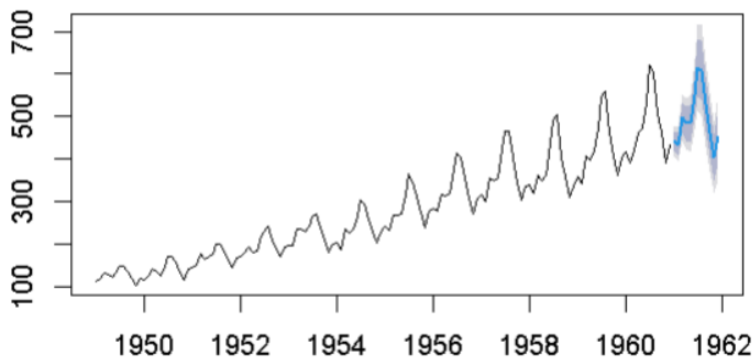
Aim: b) Time series Forecasting using Exponential smoothing, ARIMA models.

Code:

```
library(forecast)
data(AirPassengers)
ts_data <- AirPassengers
# Exponential Smoothing
fit_ets <- ets(ts_data)
forecast_ets <- forecast(fit_ets, h = 12)
plot(forecast_ets, main = "Exponential Smoothing Forecast")
# ARIMA Model
fit_arma <- auto.arima(ts_data)
forecast_arma <- forecast(fit_arma, h = 12)
plot(forecast_arma, main = "ARIMA Forecast")
```

Output:

Exponential Smoothing Forecast



ARIMA Forecast

